A ternary Möbius sum with one short variable

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Abstract

For integers $N \geq 3$ and $2 \leq H \leq N$ and complex weights u, v with $\|u\|_\infty, \|v\|_\infty \leq 1$ let $S_{u,v}(H,N) = \sum_{n_1+n_2+n_3=N,\, n_3 \leq H} u_{n_1} v_{n_2} \mu(n_1 n_2 n_3)$, the sum over ordered triples of positive integers, and let S_μ be its case $u = v = 1$. Banks and Shparlinski proved $S_{u,v}(H,N) \ll_{\varepsilon,C} NH(\log N)^{-C}$ for every $C > 0$ in the range $H \geq \exp((\log N)^\varepsilon)$, and asked how small H may be taken. We prove that for every $A > 0$ there is a constant $K(A)$ with

$$|S_{u,v}(H,N)| \;\leq\; K(A)\, NH(\log H)^{-A} + c_1 \frac{N \log N}{\log\log N}$$

for all $2 \leq H \leq N$ and all such u, v, where c_1 is absolute, so that $S_{u,v}(H,N) = o(NH)$ whenever $H \log\log N / \log N \to \infty$. For all $N \leq x$ outside $O(x/\log x)$ exceptions the second term improves to $N \log\log x$, and then $H/\log\log x \to \infty$ suffices. The proof truncates the decomposition over the common divisor e of n_1 and n_2, using that the equation forces $n_3 \equiv N \pmod e$. The analytic input is Davenport's 1937 bound for $\sum_{n \leq x} \mu(n)\mathbf{e}(\alpha n)$, uniform in α.

2020 Mathematics Subject Classification: 11N37 (primary); 11L07, 11P32 (secondary).
Keywords: Möbius function; additive problems; short variables; exponential sums.

1 Introduction

For integers $N \geq 3$ and $2 \leq H \leq N$ and complex weights $u = \{u_n\}$, $v = \{v_n\}$ with $\|u\|_\infty, \|v\|_\infty \leq 1$ put

$$S_{u,v}(H,N) \;=\; \sum_{\substack{n_1+n_2+n_3=N \\ n_3 \leq H}} u_{n_1} v_{n_2}\, \mu(n_1 n_2 n_3), \qquad S_\mu(H,N) = S_{u,v}(H,N)\big|_{u=v=1}, \tag{1}$$

the sums taken over ordered triples of positive integers; this is the generality of [1, Thm. 2.1] at $f = g = \wp = \mathrm{id}$. The number of triples is of order NH, and $|S_{u,v}(H,N)| \leq NH$ is the trivial bound.

Banks and Shparlinski [1, (1.3)] introduced $S_\mu(H,N)$ as a Möbius analogue of the ternary Goldbach problem with one variable restricted to a short range, and asked how short that range may be: "it is natural to seek nontrivial bounds for $S_\mu(H,N)$ when H is as small as possible relative to N" [1, §7.6]. Their unconditional answer, obtained from their general bilinear theorem, is that for a fixed $\varepsilon > 0$ and every $C > 0$,

$$S_\mu(H,N) \;\ll_{\varepsilon,C}\; NH(\log H)^{-C/\varepsilon} \;\leq\; NH(\log N)^{-C} \qquad \big(H \geq \exp((\log N)^\varepsilon)\big). \tag{2}$$

At the opposite end, $H = 1$ reduces $S_\mu(H,N)$ to a sum over pairs, which Banks and Shparlinski describe as a variant of Chowla's problem and out of reach of current methods [1, §7.6]. For the von Mangoldt function in place of μ the corresponding sum is positive for sufficiently large odd N once $H \geq N^{11/400+\varepsilon}$ [2], and once $H \geq N^{11/420}$ [5, Thm. 7.3].

Theorem 1. *There is an absolute constant c_1 with the following property. For every $A > 0$ there is a constant $K(A)$ such that*

$$|S_{u,v}(H,N)| \;\le\; K(A)\,NH(\log H)^{-A} + c_1\,\frac{N\log N}{\log\log N} \tag{3}$$

for every integer $N \ge 3$, every integer H with $2 \le H \le N$, and all u, v as above.

Every constant in the proof is effective apart from $K(A)$, which inherits the constant of Davenport's theorem.

Corollary 2. (a) *If $A > 0$ and $\eta > 0$ are fixed and $(\log N)^{1+\eta} \le H \le N$, then $|S_{u,v}(H,N)| \ll_{A,\eta} NH(\log\log N)^{-A}$.*

(b) *$S_{u,v}(H,N) = o(NH)$, uniformly in u and v, as $H\log\log N/\log N \to \infty$ with $H \le N$.*

(c) *Theorem 1 contains (2), for fixed $\varepsilon > 0$ and every $C > 0$, throughout $\exp((\log N)^{\varepsilon}) \le H \le N$.*

Part (b) covers H of order $\log N$, and H down to $(\log N)/(\log\log N)^{1-\delta}$ for any fixed $\delta > 0$.

Corollary 3. *There is an absolute constant c_2 with the following property. Let $A > 0$, let $x \ge 16$ and let $H \ge 2$ be an integer, and let $K(A)$ be the constant obtained in the proof of Theorem 1. For every integer N with $\max(H,3) \le N \le x$, apart from $O(x/\log x)$ exceptions depending on x and H alone,*

$$|S_{u,v}(H,N)| \;\le\; K(A)\,NH(\log H)^{-A} + c_2\,N\log\log x \tag{4}$$

for all u, v as above. If $H = H(x)$ with $H/\log\log x \to \infty$, then $S_{u,v}(H,N) = o(NH)$ as $x \to \infty$, uniformly in those N and in u, v.

The restriction $H \ge \exp((\log N)^{\varepsilon})$ in (2) comes from the sum over the common divisor of n_1 and n_2. That sum contributes a factor $\log\max\{A,B\}$ to the bound for their $K(d_1,d_2)$ preceding [1, (4.6)], which is $\log N$ in the present application, and the hypothesis of [1, Thm. 2.1] converts the factor into a power of $\log H$. Writing $\mu(n_1n_2) = \mu(n_1)\mu(n_2)\sum_{e\mid(n_1,n_2)}\mu(e)$, the available bound for the e-th term decays like $1/e$ and the sum runs to N. If the decomposition over e is performed first, the conditions $e \mid n_1$ and $e \mid n_2$ combine with $n_1 + n_2 + n_3 = N$ to give $n_3 \equiv N \pmod e$, so the short variable is restricted as well; the e-th term then involves at most $(H/e+1)(N/e)$ triples. The tail beyond E contributes $O(NH/E)$ from $e \le H$ and, from $e > H$, a quantity measured by the number of divisors of the integers $N - n_3$ with $n_3 \le H$. The analytic argument is applied only to $e \le E$, where the loss is a power of $\log E$. In the generality of [1, Thm. 2.1] the two long variables enter through arbitrary injections and the third through a bracket polynomial, so $e \mid n_1$ and $e \mid n_2$ restrict no other variable, only one factor $1/e$ is available, and the counting bound for the tail of the e-sum gives no saving there. The treatment of the terms with $e \le E$, including the second inclusion–exclusion and its truncation, follows [1, §4].

2 Notation and inputs

Throughout, p is a prime, μ, τ and ω are the Möbius function, the divisor function and the number of distinct prime factors, (m,n) is the greatest common divisor and $[m,n]$ the least common multiple, and $\mathbf{e}(x) = e^{2\pi ix}$. For a weight $w = \{w_n\}$ with $\|w\|_\infty \le 1$, for $q \ge 1$ and $\alpha \in \mathbb{R}$ set

$$A_q(w;\alpha) = \sum_{\substack{n\le N\\ q\mid n}} w_n\mu(n)\mathbf{e}(\alpha n)\,, \qquad \Psi_d(\alpha) = \sum_{\substack{m\le H\\ d\mid m}} \mu(m)\mathbf{e}(\alpha m)\,, \tag{5}$$

and, for $e \ge 1$ and $d \ge 1$,

$$\Phi_{e,d}(\alpha) = \sum_{\substack{n_1,n_2 \le N \\ e \mid n_1,\ e \mid n_2 \\ d \mid n_1 n_2}} u_{n_1} v_{n_2} \mu(n_1)\mu(n_2)\mathbf{e}(\alpha(n_1+n_2))\,. \tag{6}$$

Lemma 4. *For every $C > 0$ there is a constant $c(C)$ such that, for $H \ge 3$,*

$$\sup_{\alpha \in \mathbb{R}} \max_{r \ge 1,\ a \in \mathbb{Z}} \Big| \sum_{m \le H,\ m \equiv a\,(r)} \mu(m)\mathbf{e}(\alpha m) \Big| \ \le\ c(C)\, H (\log H)^{-C};$$

the case $(a,r) = (0,d)$ is Ψ_d.

Proof. Davenport [3] proved $\sup_{\beta \in \mathbb{R}} \big|\sum_{m \le x} \mu(m)\mathbf{e}(\beta m)\big| \ll_C x(\log x)^{-C}$ for $x \ge 3$ and every $C > 0$; see also [4, Thm. 13.10]. Detecting $m \equiv a \pmod r$ by additive characters modulo r, the sum above equals

$$\frac{1}{r} \sum_{j \bmod r} \mathbf{e}(-ja/r) \sum_{m \le H} \mu(m)\mathbf{e}((\alpha + j/r)m)\,,$$

and each inner sum is $O_C(H(\log H)^{-C})$ by Davenport's bound, uniformly in α and j. □

The same statement for a bracket polynomial in place of the linear phase is [1, Lem. 3.1], with [1, Rem. 3.2] recording the independence of the implied constant from the progression.

Lemma 5. *Let $k \ge 1$ be an integer and write $\ell = \log x$. For $x \ge 2$,*

$$\sum_{n \le x} \tau(n)^k \ll_k x\ell^{2^k-1}, \qquad \sum_{n \le x} \tau(n)^k n^{-1} \ll_k \ell^{2^k}, \qquad \sum_{n \le x} \tau(n)^k n^{-1/2} \ll_k x^{1/2}\ell^{2^k-1},$$

$$\sum_{n > x} \tau(n)^k n^{-3/2} \ll_k x^{-1/2}\ell^{2^k-1}.$$

Proof. The first estimate is [4, (1.81)]; the others follow from it by partial summation, the last two as in [1, (3.11), (3.12)]. □

Lemma 6. *Put $\tau^*(N) = \max_{1 \le m < N} \tau(m)$. There is an absolute effective constant c_0 such that*

$$\log \tau^*(N) \ \le\ c_0 \frac{\log N}{\log\log N} \qquad (N \ge 3). \tag{7}$$

Proof. Nicolas and Robin [6] give an explicit constant c with $\log \tau(m) \le c\,(\log m)/(\log\log m)$ for every $m \ge 3$. For $16 \le m < N$ the right side is at most $c\,(\log N)/(\log\log N)$, since $t \mapsto (\log t)/(\log\log t)$ increases for $t \ge e^e$; and for $m < 16$ one has $\tau(m) \le 6$ while $(\log N)/(\log\log N) \ge e$ for $N \ge 3$. So (7) holds with $c_0 = \max(c, 1)$. □

We also use, for squarefree e, $x \ge 3$ and $r \ge 1$,

$$\sum_{t \mid e} \tau(t) = 3^{\omega(e)}, \qquad \sum_{t \mid e} \tau(t)^2 = 5^{\omega(e)}, \qquad \sum_{t \mid e} t\,\tau(t) = \prod_{p \mid e}(1+2p) \le 3^{\omega(e)} e, \tag{8}$$

$$\sum_{e \le x} \mu^2(e)\, r^{\omega(e)} e^{-1} \le \prod_{p \le x} \Big(1 + \frac{r}{p}\Big) \ \ll_r\ (\log x)^r, \tag{9}$$

the second line being Mertens' theorem.

3 The decomposition and its tail

Lemma 7. *For $e \geq 1$ put*

$$W(e) = \sum_{\substack{n_1+n_2+n_3=N \\ n_3 \leq H \\ e \mid n_1,\ e \mid n_2}} u_{n_1} v_{n_2} \mu(n_1)\mu(n_2)\mu(n_3) \big[(n_1 n_2, n_3) = 1\big], \tag{10}$$

where $[\,\cdot\,]$ is 1 if the condition holds and 0 otherwise. Then $S_{u,v}(H,N) = \sum_{e \leq N} \mu(e) W(e)$.

Proof. For positive integers n_1, n_2, n_3,

$$\mu(n_1 n_2 n_3) = \mu(n_1)\mu(n_2)\mu(n_3) \big[(n_1, n_2) = 1\big] \big[(n_1 n_2, n_3) = 1\big]. \tag{11}$$

If two of the three share a prime factor then $n_1 n_2 n_3$ is not squarefree and both sides vanish; otherwise n_1, n_2, n_3 are pairwise coprime and (11) is multiplicativity. Multiply (11) by $u_{n_1} v_{n_2}$, insert $[(n_1, n_2) = 1] = \sum_{e \mid (n_1, n_2)} \mu(e)$ and exchange the two finite sums; since $n_1 \leq N$, only $e \leq N$ occurs. □

Lemma 8. *For every $e \geq 1$,*

$$|W(e)| \;\leq\; \Big(\Big\lfloor \frac{H}{e} \Big\rfloor + 1\Big) \Big\lfloor \frac{N}{e} \Big\rfloor, \tag{12}$$

and, with

$$X_N = \sum_{1 \leq n_3 \leq H} \tau(N - n_3) \quad (\tau(m) = 0 \text{ for } m \leq 0), \qquad \text{so that} \quad H \leq X_N \leq H\tau^*(N), \tag{13}$$

for every integer $E \geq 1$,

$$\sum_{E < e \leq N} |W(e)| \;\leq\; \frac{2NH}{E} + N \log \frac{X_N}{H} + N. \tag{14}$$

Proof. If $e \mid n_1$, $e \mid n_2$ and $n_1 + n_2 + n_3 = N$ then $n_3 \equiv N \pmod e$, so n_3 takes at most $\lfloor H/e \rfloor + 1$ values in $[1, H]$. For each of them n_1 runs through multiples of e in $[1, N]$, at most $\lfloor N/e \rfloor$ values, and $n_2 = N - n_3 - n_1$ is determined. Every summand in (10) has modulus at most 1, which gives (12).

The upper bound in (13) holds because $\tau(N - n_3) \leq \tau^*(N)$ for $n_3 < N$, and the lower bound because each such term is at least 1, the case $H = N$ giving $X_N = \sum_{m<N} \tau(m) \geq 2N - 3 \geq N$. Write $X = X_N$ and split the range of (14) at H and at X, any of the three parts being possibly empty. If $E < e \leq H$ then $\lfloor H/e \rfloor + 1 \leq 2H/e$, so by (12) the e-th term is at most $2NH/e^2$, and $\sum_{e>E} 2NH/e^2 \leq 2NH/E$ because E is an integer. If $H < e \leq X$ then $\lfloor H/e \rfloor = 0$ and (12) gives $|W(e)| \leq N/e$, so these e contribute at most $N \sum_{H<e\leq X} 1/e \leq N \log(X/H)$. Finally let $e > X$. A triple counted in (10) has $n_1 + n_2 = N - n_3 \geq 2$ with $e \mid n_1$ and $e \mid n_2$, so $e \mid N - n_3$; hence $W(e) = 0$ unless e divides one of the integers $N - n_3$ with $n_3 \leq H$ and $n_3 < N$, and $|W(e)| \leq N/e < N/X$ otherwise. The number of pairs (n_3, e) with $n_3 \leq H$, $n_3 < N$ and $e \mid N - n_3$ is X, so the third part contributes at most $X \cdot N/X = N$. □

Remark 9. Counting the divisors $e > H$ of (14) dyadically with $Y_j = 2^j H$ gives

$$\sum_{H < e \leq N} |W(e)| \;\leq\; N \sum_{j \geq 0} \min\Big(1,\; \frac{1}{Y_j} \sum_{n_3 \leq H,\ n_3 < N} \#\{e \mid N - n_3 :\ Y_j < e \leq 2Y_j\}\Big),$$

since a block holds at most Y_j integers and $|W(e)| \leq N/Y_j$ on it. The first entry for $Y_j \leq X_N$ and the second above it recover (14) up to a constant; using $|W(e)| \leq N/e$ directly instead gives only $N \sum_{H<e\leq N} e^{-1} \leq N \log(N/H)$.

4 The bilinear factor and the large divisors

Lemma 10. *Let $e \ge 1$ and let $d \ge 1$ be squarefree. Then*

$$\int_0^1 |\Phi_{e,d}(\alpha)|\, d\alpha \;\le\; \tau(d)^2\, \frac{N\,(e,d)^{1/2}}{e\,d^{1/2}}.$$

Proof. Let n_1, n_2 be positive integers. If $d \mid n_1 n_2$, put $d_1 = \prod_{p \mid d,\ p \mid n_1} p$ and $d_2 = d/d_1$. Then $d = d_1 d_2$ with $(d_1, d_2) = 1$, and $d_1 \mid n_1$ because d is squarefree, while every prime of d_2 divides n_2 and not n_1, so $d_2 \mid n_2$ and $(n_1, d_2) = 1$. Conversely those conditions give $d \mid n_1 n_2$ and recover the same pair (d_1, d_2), since a prime of d divides n_1 if and only if it divides d_1. Hence

$$\big[d \mid n_1 n_2\big] = \sum_{\substack{d_1 d_2 = d \\ (d_1, d_2) = 1}} \big[d_1 \mid n_1\big]\big[d_2 \mid n_2\big]\big[(n_1, d_2) = 1\big]. \tag{15}$$

Writing $[(n_1, d_2) = 1] = \sum_{k \mid (n_1, d_2)} \mu(k)$ and inserting (15) into (6), the conditions on n_1 become $[e, d_1, k] \mid n_1$ and those on n_2 become $[e, d_2] \mid n_2$, so that

$$\Phi_{e,d}(\alpha) = \sum_{\substack{d_1 d_2 = d \\ (d_1, d_2) = 1}} \sum_{k \mid d_2} \mu(k)\, A_{[e,d_1,k]}(u;\alpha)\, A_{[e,d_2]}(v;\alpha).$$

By Cauchy–Schwarz and Parseval, for any $q_1, q_2 \ge 1$ and any weights w, w' bounded by 1,

$$\int_0^1 |A_{q_1}(w;\cdot) A_{q_2}(w';\cdot)|\, d\alpha \le \|A_{q_1}(w;\cdot)\|_2 \|A_{q_2}(w';\cdot)\|_2, \qquad \|A_q(w;\cdot)\|_2^2 = \sum_{\substack{n \le N \\ q \mid n}} |w_n|^2 \mu^2(n) \le \frac{N}{q},$$

so each term contributes at most $N(q_1 q_2)^{-1/2}$ with $q_1 = [e, d_1, k]$ and $q_2 = [e, d_2]$. Since $(d_1, d_2) = 1$ and d is squarefree, $(e, d_1)(e, d_2) = (e, d)$, whence

$$q_1 q_2 \;\ge\; [e, d_1][e, d_2] = \frac{e^2 d_1 d_2}{(e, d_1)(e, d_2)} = \frac{e^2 d}{(e, d)}.$$

There are at most $\tau(d)$ factorisations $d = d_1 d_2$ and at most $\tau(d)$ values of k. □

Lemma 11. *Let $e \ge 1$, let $d \ge 1$ be squarefree, and put*

$$I(e,d) = \sum_{\substack{n_1 + n_2 + n_3 = N,\ n_3 \le H \\ d \mid n_3,\ d \mid n_1 n_2 \\ e \mid n_1,\ e \mid n_2}} u_{n_1} v_{n_2} \mu(n_1)\mu(n_2)\mu(n_3). \tag{16}$$

Then

$$|I(e,d)| \;\le\; \tau(d)\Big(\frac{H(d,e)}{de} + 1\Big)\Big(\frac{N(e,d)}{e\,d^{1/2}} + 1\Big).$$

Proof. Every summand in (16) has modulus at most 1, so it is enough to count the triples. As in Lemma 8, $n_3 \equiv N \pmod e$; with $d \mid n_3$ this confines n_3 to a single residue class modulo $[d, e] = de/(d, e)$ when $(d, e) \mid N$, and admits no n_3 otherwise. So n_3 takes at most $H(d, e)/(de) + 1$ values in $[1, H]$.

Fix such an n_3 and let $d = d_1 d_2$ be the factorisation of (15) attached to the pair (n_1, n_2); there are at most $\tau(d)$ possibilities for it, and we bound the number of n_1 for each. Put $d_0 = \max(d_1, d_2) \ge d^{1/2}$. If $d_0 = d_1$ then $d_0 \mid n_1$, and if $d_0 = d_2$ then $d_0 \mid n_2 = N - n_3 - n_1$, so that $n_1 \equiv N - n_3 \pmod{d_0}$. In either case n_1 lies in one residue class modulo d_0 and in one modulo e, hence in at most $N(e, d_0)/(ed_0) + 1 \le N(e, d)/(ed^{1/2}) + 1$ values of $[1, N]$, since $d_0 \mid d$ and $d_0 \ge d^{1/2}$; and n_2 is then determined. □

5 Proofs of Theorem 1 and Corollaries 2, 3

Proof of Theorem 1. Fix $A > 0$ and set

$$C = 2A + 4, \qquad E = \lceil (\log H)^{A+1} \rceil, \qquad D = \lceil (\log H)^{2A+6} \rceil. \tag{17}$$

Let $H_1(A)$ and $N_1(A)$ be effective thresholds, at our disposal, and assume $H \geq H_1(A)$ and $N \geq N_1(A)$; taking $H_1(A)$ large, we also have $2E \leq D \leq H$. For $2 \leq H < H_1(A)$ the trivial bound gives $|S_{u,v}(H,N)| \leq NH \leq (\log H_1(A))^A \, NH(\log H)^{-A}$, and the range $N < N_1(A)$ is covered the same way, so both are absorbed by enlarging $K(A)$. By Lemma 7 and Lemma 8, and since $|\mu(e)| \leq 1$ with $\mu(e) = 0$ unless e is squarefree,

$$|S_{u,v}(H,N)| \;\leq\; \sum_{\substack{e \leq E \\ \mu^2(e)=1}} |W(e)| \;+\; \frac{2NH}{E} + N \log \frac{X_N}{H} + N. \tag{18}$$

Fix a squarefree $e \leq E$. Since $d \mid n_3 \leq H$, inserting $[(n_1 n_2, n_3) = 1] = \sum_{d \mid (n_1 n_2, n_3)} \mu(d)$ into (10) gives $W(e) = \sum_{d \leq H} \mu(d) I(e,d)$, with $I(e,d)$ as in (16) and only squarefree d contributing. By orthogonality,

$$I(e,d) = \int_0^1 \Phi_{e,d}(\alpha) \Psi_d(\alpha) \mathbf{e}(-\alpha N) \, d\alpha, \tag{19}$$

the integral selecting from the ranges in (5) and (6) the triples with $n_1 + n_2 + n_3 = N$.

Divisors $d \leq D$. By (19), Lemma 4 and Lemma 10,

$$\sum_{\substack{d \leq D \\ \mu^2(d)=1}} |I(e,d)| \;\leq\; c(C) H (\log H)^{-C} \cdot \frac{N}{e} \cdot \sum_{\substack{d \leq D \\ \mu^2(d)=1}} \tau(d)^2 (e,d)^{1/2} d^{-1/2}.$$

Write $t = (e,d)$ and $d = td'$, and use $\tau(td') \leq \tau(t)\tau(d')$ together with Lemma 5 at $k = 2$, which applies because $t \leq e \leq E \leq 2(\log H)^{A+1}$ and hence $D/t \geq (\log H)^{A+5}/2 \geq 2$:

$$\sum_{d \leq D} \tau(d)^2 (e,d)^{1/2} d^{-1/2} \leq \sum_{t \mid e} \tau(t)^2 \sum_{d' \leq D/t} \tau(d')^2 d'^{-1/2} \;\ll\; D^{1/2} (\log D)^3 \sum_{t \mid e} \tau(t)^2,$$

which is $\ll 5^{\omega(e)} D^{1/2} (\log D)^3$ by (8). Summing over e and using (9) with $r = 5$,

$$\sum_{\substack{e \leq E \\ \mu^2(e)=1}} \sum_{\substack{d \leq D \\ \mu^2(d)=1}} |I(e,d)| \;\ll\; c(C) \, NH (\log H)^{-C} D^{1/2} (\log D)^3 (\log E)^5. \tag{20}$$

Divisors $d > D$. Expand the product in Lemma 11 into four terms. With $t = (e,d)$ as above, and with D/t and H/t again at least 2, Lemma 5 at $k = 1$ and (8) give

$$\begin{aligned}
\sum_{d > D} \tau(d) (e,d)^2 d^{-3/2} &\leq \sum_{t \mid e} t^{1/2} \tau(t) \sum_{d' > D/t} \tau(d') d'^{-3/2} \;\ll\; 3^{\omega(e)} e \, D^{-1/2} \log D, \\
\sum_{d \leq H} \tau(d) (e,d) d^{-1/2} &\;\ll\; 3^{\omega(e)} H^{1/2} \log H, \\
\sum_{d \leq H} \tau(d) (e,d) d^{-1} &\;\ll\; 3^{\omega(e)} (\log H)^2, \qquad \sum_{d \leq H} \tau(d) \;\ll\; H \log H.
\end{aligned}$$

Multiplying by the prefactors NH/e^2, N/e, H/e and 1 of the four terms and summing over squarefree $e \leq E$ with (9) at $r = 3$,

$$\begin{aligned}
&\sum_{\substack{e \leq E \\ \mu^2(e)=1}} \sum_{\substack{D < d \leq H \\ \mu^2(d)=1}} |I(e,d)| \\
&\ll\; NHD^{-1/2} (\log D)(\log E)^3 + NH^{1/2} (\log H)(\log E)^3 + H(\log H)^2 (\log E)^3 + EH \log H.
\end{aligned} \tag{21}$$

Collecting. Insert (20) and (21) into (18) and use (17). By $X_N \le H\tau^*(N)$, (7) and $(\log N)/(\log\log N) \ge e$, the terms of (18) other than the sum are at most $2NH(\log H)^{-A-1} + (c_0+1)N(\log N)/(\log\log N)$. In (20), $D^{1/2} \le 2(\log H)^{A+3}$ and $C = 2A+4$ leave $\ll c(C)NH(\log H)^{-A-1}(\log\log H)^8$, which is $\ll c(C)NH(\log H)^{-A}$ once $(\log\log H)^8 \le \log H$. In (21) the first term is $\ll NH(\log H)^{-A-3}(\log\log H)^4$; the second is $\ll NH(\log H)^{-A}$ once $H^{1/2} \ge (\log H)^{A+1}(\log\log H)^4$; and the last two are $\ll H(\log H)^{A+2}(\log\log H)^3$, which is $\ll NH(\log H)^{-A}$ because $(\log H)^{2A+2}(\log\log H)^3$ is at most $(\log N)^{2A+2}(\log\log N)^3$, and that is $\le N$ for $N \ge N_1(A)$. Each condition here involves A alone, which fixes $H_1(A)$ and $N_1(A)$, and every implied constant depends on A only. □

Remark 12. The weights enter only through $|u|, |v| \le 1$, in the counting bounds (12) and Lemma 11 and in Parseval's identity for (5). The bound also holds with n_3 confined to a progression $n_3 \equiv a \pmod q$, uniformly in a and q: that congruence together with $d \mid n_3$ is one class modulo $[d,q]$ or none, so Lemma 4 covers the resulting Ψ_d, while in (12), (14) and Lemma 11 it only reduces the number of admissible n_3.

Proof of Corollary 2. (a) Apply Theorem 1 with the same A. Here $(\log H)^{-A} \le \big((1+\eta)\log\log N\big)^{-A}$, while $N(\log N)/(\log\log N) \le NH(\log N)^{-\eta}$, which is $\le NH(\log\log N)^{-A}$ for N large.

(b) Put $\rho = H\log\log N/\log N$ and apply Theorem 1 with $A=1$, giving $K(1)NH(\log H)^{-1} + c_1N(\log N)/(\log\log N)$. The second summand is $c_1NH/\rho = o(NH)$. For the first, $H = \rho(\log N)/(\log\log N) \ge e\rho \to \infty$, since $(\log N)/(\log\log N) \ge e$ for $N \ge 3$, so $(\log H)^{-1} \to 0$.

(c) Given $C > 0$ apply Theorem 1 with $A = C/\varepsilon$. Then $(\log H)^{-C/\varepsilon} \le (\log N)^{-C}$, while $N(\log N)/(\log\log N) \le N\log N \le NH(\log H)^{-C/\varepsilon}$ for N large, because $t \mapsto t(\log t)^{-C/\varepsilon}$ increases for $\log t \ge C/\varepsilon$ and its value at $t = \exp((\log N)^{\varepsilon})$ is $\exp((\log N)^{\varepsilon})(\log N)^{-C} \ge \log N$. □

Proof of Corollary 3. Let $\mathfrak{E} = \{N \le x : X_N > H(\log x)^2\}$. By (13) and Lemma 5 at $k=1$,

$$\sum_{N\le x} X_N = \sum_{n_3\le H}\ \sum_{N\le x} \tau(N-n_3) \le H\sum_{m\le x}\tau(m) \ll Hx\log x,$$

so $|\mathfrak{E}| \ll x\log x/(\log x)^2 = x/\log x$, with an absolute implied constant. For $N \notin \mathfrak{E}$ one has $\log(X_N/H) \le 2\log\log x$, so the proof of Theorem 1 applies with $3N\log\log x$ in place of the last two terms of (18), using $\log\log x \ge 1$; this gives (4). If $H/\log\log x \to \infty$ then $H \to \infty$, and (4) reads $NH\big(K(A)(\log H)^{-A} + c_2(\log\log x)/H\big)$ with both summands tending to 0. □

Formalisation

The exact finite steps—the factorisation (11), Lemma 7, the bounds (12), (13) and (14) of Lemma 8, the splitting (15), and the residue-class count behind Lemma 11—are verified in Lean 4 with mathlib, in `P10-short-variable-mobius/lean/` of the public repository https://github.com/soke91/arithmetic-convolution-fields. The analytic estimates and Theorem 1 itself are not formalised.

Funding

No external funding was received for this research.

Acknowledgements and AI use disclosure

Claude Opus 5.5 (Anthropic) and ChatGPT (OpenAI; GPT-6) were used in developing, checking and writing the mathematical arguments and in preparing the manuscript. The author directed the work, reviewed it, and takes responsibility for its contents.

Declaration of competing interests

The author declares no competing interests.

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